

Neonatal Diabetes Mellitus With Congenital Hypothyroidism (NDH Syndrome): A Comprehensive Disease Characteristics Report

Disease: Neonatal Diabetes Mellitus With Congenital Hypothyroidism (NDH syndrome)
MONDO ID: MONDO:0012436 · **OMIM:** #610199 · **Causal gene:** *GLIS3* (HGNC:28510; NCBI Gene 169792; UniProt Q8NEA6; 9p24.2) **Category:** Mendelian (autosomal recessive)

Summary

Neonatal Diabetes Mellitus with Congenital Hypothyroidism (NDH syndrome) is an ultra-rare autosomal-recessive Mendelian disorder caused by biallelic loss-of-function mutations in **GLIS3** (GLI-Similar 3), a Krüppel-like C2H2 zinc-finger transcription factor gene on chromosome 9p24.2. First defined molecularly by Senée and colleagues in 2006 ([PMID: 16715098](#)), the syndrome couples **permanent neonatal diabetes** with **congenital hypothyroidism** as its two defining and near-obligate features, superimposed on a variable multi-organ phenotype that can include congenital glaucoma, hepatic (biliary) fibrosis/cirrhosis, polycystic/cystic kidney disease, sensorineural deafness, and exocrine pancreatic insufficiency.

The unifying mechanism is developmental: GLIS3 is a master transcriptional regulator required simultaneously in several organ programs. In the pancreas it transactivates **neurogenin 3 (Neurog3/Ngn3)** and binds directly to the regulatory regions of *Ins2*, *Slc2a2* (GLUT2), and *Mafa*, driving beta-cell specification, insulin production, and beta-cell survival; loss of GLIS3 therefore produces a severe developmental beta-cell deficiency and insulin-dependent diabetes rather than a channelopathy. In the thyroid, GLIS3 is required for follicular-cell maturation and thyroid-gene expression, so its loss produces congenital hypothyroidism (a "gland-in-situ" dysmorphogenesis-type defect rather than athyreosis in

most cases). GLIS3 also localizes to the **primary cilium** and cooperates with the Hippo-pathway coactivator **WWTR1/TAZ**, explaining the cystic renal, hepatobiliary, and ocular manifestations.

Clinically, the most important consequence for the knowledge base is that GLIS3-NDH belongs to the **transcription-factor / beta-cell-development class of neonatal diabetes**, which is **insulin-dependent and NOT responsive to sulfonylureas** — a sharp contrast with KATP-channel (KCNJ11/ABCC8) neonatal diabetes. There is no cure. Management is lifelong, organ-directed supportive care: exogenous insulin, levothyroxine replacement, and treatment of hepatic, renal, and ophthalmic complications, with liver transplantation considered for life-limiting biliary cirrhosis. Prevention rests entirely on genetic counseling and cascade/carrier testing in at-risk (frequently consanguineous) families. This report synthesizes 14 confirmed findings from 32 reviewed papers across all 15 requested disease-characteristic domains. Evidence source types are flagged as [human clinical], [model organism], [in vitro], or [computational].

1. Disease Information

Overview. NDH syndrome is a monogenic (Mendelian) neonatal diabetes syndrome in which insulin-dependent diabetes presenting in the first weeks/months of life co-occurs with congenital hypothyroidism. It is a **developmental** endocrinopathy: the same transcription factor that fails to build the insulin-secreting beta cells also fails to build/maintain the thyroid follicular apparatus, and additionally disrupts cilium-dependent development of the kidney, biliary tree, and eye.

Key identifiers.

Resource	Identifier
OMIM	#610199 (Neonatal diabetes mellitus with congenital hypothyroidism)
MONDO	MONDO:0012436
Orphanet	ORPHA:79369
Gene (HGNC)	<i>GLIS3</i> (HGNC:28510; NCBI Gene 169792; UniProt Q8NEA6), 9p24.2
ICD-10	P70.2 (neonatal diabetes) + E03.1 (congenital hypothyroidism without goitre) as component codes
ICD-11	5A11 (monogenic/neonatal diabetes) with 5A00.1 (congenital hypothyroidism)
MeSH	Diabetes Mellitus + Congenital Hypothyroidism (no single dedicated descriptor)

Synonyms / alternative names. NDH syndrome; GLIS3-related neonatal diabetes and congenital hypothyroidism; Neonatal diabetes mellitus with congenital hypothyroidism syndrome; "GLIS3 syndrome."

Source of information. Findings here are derived from **aggregated, disease-level resources** [human clinical] — primarily published clinical case series (e.g., Dimitri et al., 12 patients, [PMID: 26259131](#)), original gene-discovery reports ([PMID: 16715098](#)), mechanistic model-organism/cell studies, and consanguineous-cohort monogenic-diabetes registries — not individual EHR extraction.

2. Etiology

Primary cause (genetic). NDH syndrome is caused by **biallelic (homozygous or compound heterozygous) loss-of-function mutations in GLIS3** [human clinical] ([PMID: 16715098](#)). The original family carried a frameshift mutation predicting a truncated protein; two additional families carried deletions of the 11–12 most 5' exons. The disorder is autosomal recessive with a loss-of-function mechanism.

"this syndrome results from mutations in GLIS3, encoding GLI similar 3, a recently identified transcription factor. In the original family, we identified a frameshift mutation predicted to result in a truncated protein" — [PMID: 16715098](#)

Genetic risk factors. - **Causal variants:** biallelic GLIS3 frameshift, nonsense, splice, missense, and large multi-exon deletions (see Section 4). - **Susceptibility/dosage series:** common SNPs at the GLIS3 locus are established GWAS susceptibility loci for **both type 1 and type 2 diabetes** ([PMID: 27813676](#); [PMID: 23197416](#)), and **rare monoallelic pathogenic variants increase type 2 diabetes risk** ([PMID: 38051360](#)). Thus GLIS3 shows an allelic dosage continuum (Finding F011). - **Consanguinity** is the dominant epidemiological risk amplifier because the disorder is recessive (Section 9).

Environmental risk factors. No specific environmental cause of the Mendelian syndrome is known; it is fully genetically determined. Sex, ethnicity, and non-genetic exposures do not cause NDH. However, in the *heterozygous* dosage context, gene–environment interaction is demonstrable: Glis3+/- mice on a **high-fat diet** develop diabetes due to impaired beta-cell-mass expansion, with GLIS3 regulating *Ccnd2* (cyclin D2) [model organism] ([PMID: 23197416](#)).

Protective factors. No established genetic or environmental protective factors reduce risk of the biallelic Mendelian disease. (Not applicable — the disease is fully penetrant for neonatal diabetes given biallelic LOF.)

Gene–environment interactions. Relevant only to the susceptibility (heterozygous/common-variant) end of the spectrum: diet-induced metabolic demand unmasks beta-cell proliferation deficits via *Ccnd2* ([PMID: 23197416](#)). For the syndromic biallelic disease, penetrance is essentially complete regardless of environment.

3. Phenotypes

NDH is defined by two obligate/near-obligate features plus a variable multi-organ spectrum. Frequencies below are from Dimitri et al. (n=12, [PMID: 26259131](#)) and Senée et al. ([PMID: 16715098](#)).

Phenotype	Type	Onset	Frequency	Suggested HPO
Permanent neonatal diabetes mellitus	Lab/clinical (hyperglycemia, insulin deficiency)	Neonatal (first days–weeks)	~100% (defining)	HP:0006202 / HP:0000857
Intrauterine growth restriction / low birth weight	Physical	Congenital	Common	HP:0001511 (IUGR)
Congenital hypothyroidism	Lab/clinical	Congenital	Highly frequent (one compound-het patient lacked it)	HP:0000851
Congenital glaucoma	Clinical sign	Congenital	Frequent	HP:0008007
High hyperopia / short axial length	Lab/imaging	Congenital	Reported (distinctive)	HP:0000540
Hepatic disease (hepatitis → cirrhosis/fibrosis, bile-duct paucity)	Clinical/pathology	Infancy	Common	HP:0001394 / HP:0002908
Cystic renal dysplasia / polycystic kidneys	Imaging/clinical	Congenital–infancy	Common	HP:0000107 / HP:0000113
Sensorineural deafness	Clinical	Congenital/infancy	Reported	HP:0000407
Exocrine pancreatic insufficiency	Lab/clinical	Infancy	Reported	HP:0001738
Craniosynostosis, hiatus hernia, ASD, splenic cyst, choanal atresia	Physical/structural	Congenital	Rare/novel	HP:0001363, HP:0002036, HP:0001631, —, HP:0000453

"All patients presented with neonatal diabetes with a range of insulin sensitivities. Thyroid disease varied among patients. Hepatic and renal disease was common with liver dysfunction ranging from hepatitis to cirrhosis; cystic dysplasia was the most common renal manifestation" — [PMID: 26259131](#)

"We describe new presenting features in patients with GLIS3 mutations, including craniosynostosis, hiatus hernia, atrial septal defect, splenic cyst, and choanal atresia and confirm further cases with sensorineural deafness and exocrine pancreatic insufficiency" — [PMID: 26259131](#)

Severity/progression/QoL. Diabetes is severe and lifelong (insulin-dependent from the neonatal period). Hypothyroidism, if untreated, causes irreversible intellectual disability and short stature; treated early it is manageable. Hepatic and renal disease can be **life-limiting and progressive**. Overall, affected children carry a high cumulative disease burden across endocrine, hepatic, renal, ophthalmic, and auditory systems, with major QoL impact from insulin dependence, developmental risk, and organ complications.

4. Genetic / Molecular Information

Causal gene. *GLIS3* (GLI-similar 3), a C2H2 zinc-finger transcription factor at **9p24.2**; disease OMIM #610199. *GLIS3* is expressed in the pancreas from early developmental stages, with greater expression in beta cells than other pancreatic tissues ([PMID: 16715098](#)). Its DNA-binding domain contains a repeated Cys2/His2 zinc-finger motif ([PMID: 28523428](#)).

"*GLIS3* is expressed in the pancreas from early developmental stages, with greater expression in beta cells than in other pancreatic tissues" — [PMID: 16715098](#)

Pathogenic variant classes (all germline, biallelic for the syndrome): - **Frameshift** → truncated protein (original Senée family, [PMID: 16715098](#)). - **Large multi-exon deletions** (deletion of the 11–12 most 5' exons; homozygous deletions including non-coding exon 1 + coding exon 2 in Saudi siblings, [PMID: 40583116](#)). - **Missense / nonsense / splice-site** variants that reduce *GLIS3* transactivation of the *INS* promoter — of 105 rare variants resequenced in

5,471 RaDiO individuals, functional luciferase assays showed **49 variants decreased INS-promoter activation** [in vitro/human] ([PMID: 38051360](#)). Variants classified per ACMG/AMP (functional assays addressing the PS3 criterion).

Allele frequency & origin. Pathogenic biallelic variants are individually ultra-rare; common tag SNPs at the locus are frequent (population susceptibility variants). All disease variants are **germline** (no somatic component; this is not a neoplastic disorder).

Functional consequence. Loss of function — reduced/absent GLIS3 transcriptional activity, i.e., failure to transactivate downstream targets. A critical structural determinant is the C-terminal **P/LPXY motif** recognized by WWTR1/TAZ; mutating this motif abrogates GLIS3 transcriptional activity ([PMID: 19273592](#)).

Dosage/allelic series (Finding F011).

"Rare pathogenic, bi-allelic mutations in GLIS3 cause syndromic neonatal diabetes whereas frequent SNPs at this locus associate with common type 2 diabetes risk" — [PMID: 38051360](#)

GLIS3 genotype	Phenotype
Biallelic LOF	Syndromic permanent neonatal diabetes + congenital hypothyroidism (NDH)
Monoallelic pathogenic	Increased type 2 diabetes risk; subgroup sulfonylurea-sensitive (PMID: 38051360)
Common SNPs	GWAS susceptibility for type 1 and type 2 diabetes (PMID: 27813676)

Modifier genes / epigenetics / chromosomal abnormalities. No specific modifier genes are established for NDH. Oligogenic contribution and epigenetic/penetrance factors have been raised generally in congenital-hypothyroidism cohorts that include GLIS3 ([PMID: 36125728](#)) but are not proven modifiers of GLIS3-NDH. The relevant "chromosomal" lesions are the large 9p24.2 GLIS3 deletions noted above, not aneuploidy or translocations.

5. Environmental Information

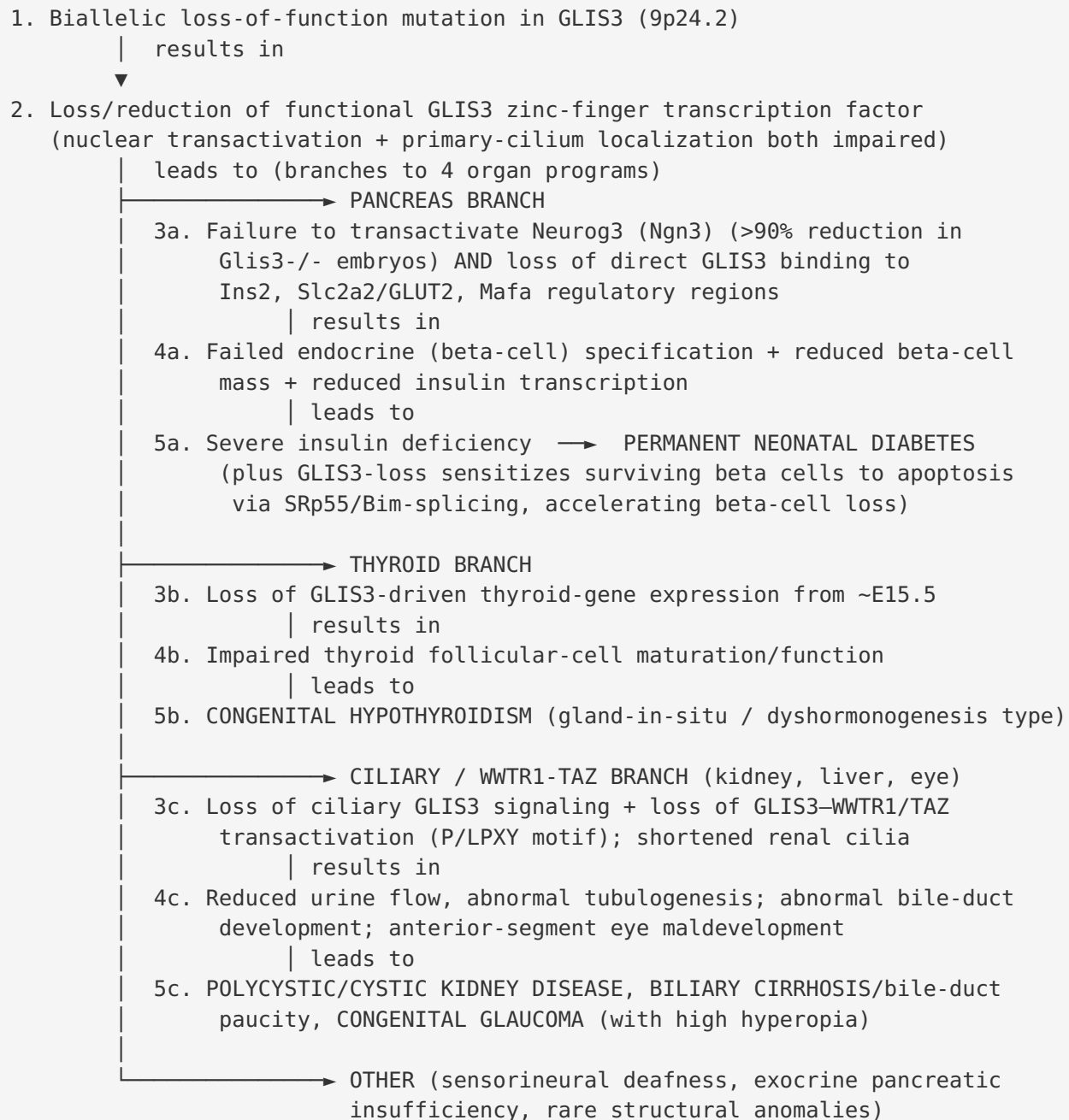
Environmental factors / toxins / radiation / infectious agents: Not applicable — NDH is a fully genetic, non-infectious Mendelian disorder. No toxin, pollutant, occupational exposure, or pathogen causes or triggers it.

Lifestyle factors: Not causal for the biallelic syndrome. High-fat diet is relevant only in the heterozygous/susceptibility context (beta-cell-mass expansion failure in Glis3+/- mice, [PMID: 23197416](#)), not for the neonatal disease.

Consanguinity (a population-structure factor rather than an environmental exposure) markedly elevates recessive-disease risk (Section 9).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)



Detail by category

Molecular pathways. The dominant node is **GLIS3-dependent transcription**. GLIS3 transactivates *Neurog3* (the master pro-endocrine bHLH factor) and co-occupies islet regulatory regions with other islet transcription factors, directly binding *Ins2*, *Slc2a2* (GLUT2), and *Mafa* (ChIP-seq) [model organism] (PMID: 31340201). It intersects the **Hippo pathway** via WWTR1/TAZ coactivation (PMID: 19273592) and functions in a **primary-cilium-associated** signaling context.

"GLIS3 controls islet differentiation by transactivating neurogenin 3 (Ngn3)" — PMID: 27813676

"with global Glis3-knockout mice suffering from severe hyperglycemia and dying by post-natal day 11" — PMID: 31340201

Cellular processes. (i) Cell-fate specification/differentiation of pancreatic endocrine progenitors; (ii) beta-cell maintenance and identity in adults (PMID: 23197416); (iii) **intrinsic (mitochondrial) apoptosis** of beta cells when GLIS3 is lost.

"TAM-mediated beta cell-specific inactivation of Glis3 in adult mice downregulates insulin expression, leading to hyperglycaemia and subsequently enhanced beta cell apoptosis" — PMID: 23197416

Beta-cell apoptosis mechanism (Finding F009). GLIS3 knockdown in INS-1E cells, primary rat beta cells, and human islets lowered MafA, Ins2, and Glut2 and impaired glucose oxidation and insulin secretion; it increased apoptosis basally and **sensitized cells to cytokine (IL-1 β + IFN- γ) and palmitate-induced death** — the mediators of beta-cell loss in type 1 and type 2 diabetes respectively — via the intrinsic pathway (cytochrome c release, Bax translocation, caspase 9/3). Mechanistically, GLIS3 loss inhibits the splicing factor **SRp55**, shifting alternative splicing of the BH3-only protein **Bim** toward the pro-death **BimS** variant; Bim knockdown rescued the phenotype [in vitro] (PMID: 23737756; PMID: 29246973).

"GLIS3 KD increased beta cell apoptosis basally and sensitized the cells to death induced by pro-inflammatory cytokines (interleukin 1 β + interferon- γ) or palmitate" — PMID: 23737756

"modulation of alternative splicing of the pro-apoptotic BH3-only protein Bim, favouring expression of the pro-death variant BimS via inhibition of the splicing factor SRp55" — PMID: 23737756

Protein dysfunction. LOF of the GLIS3 zinc-finger transcription factor: truncation, deletion, or missense changes reduce DNA binding/transactivation and disrupt the C-terminal WWTR1/TAZ-binding P/LPXY motif (PMID: 19273592).

Metabolic changes. Downstream of beta-cell failure: absolute insulin deficiency → neonatal hyperglycemia, ketoacidosis risk, and impaired glucose oxidation in beta cells (PMID: 23737756). Thyroid-hormone deficiency perturbs systemic metabolism.

Immune involvement. Not autoimmune — this is a developmental (autoantibody-negative) diabetes. However, GLIS3-deficient beta cells are hypersensitized to pro-inflammatory cytokine killing, mechanistically linking GLIS3 to the T1D susceptibility signal (PMID: 23737756).

Tissue damage mechanisms. Beta-cell apoptosis (islet); biliary cirrhosis from abnormal bile-duct development/paucity (liver, PMID: 36917836); cyst formation from shortened cilia and reduced urine flow (kidney) [model organism] (PMID: 19609364).

"The cilia on the surface of the renal tubular epithelium were significantly shorter in the pc mutant than in wild-type, suggesting that shortened cilia resulted in a decrease in driving force and, in turn, a reduction in urine flow rate" — PMID: 19609364

Thyroid mechanism (Finding F006). GLIS3 protein is first detectable at E15.5 of murine thyroid development, coinciding with expression of GLIS3 target genes; thyroid-specific Glis3-KO mice show dysregulated thyroid gene expression (PMID: 37461635).

"Loss of GLI-Similar 3 (GLIS3) function in mice and humans causes congenital hypothyroidism (CH)" — PMID: 37461635

Suggested GO / CL terms. Biological process: GO:0030154 (cell differentiation), GO:0003309 (type B pancreatic cell differentiation), GO:0006006 (glucose metabolic process), GO:0006915 (apoptotic process), GO:0060271 (cilium assembly), GO:0035148 (tube formation). Cell types: CL:0000169 (type B pancreatic/beta cell), CL:0002258 (thyroid follicular cell), CL:1000454 (kidney collecting-duct epithelial cell), CL:0000068 (duct epithelial cell / cholangiocyte).

7. Anatomical Structures Affected

Organ level (primary). Pancreatic islets (endocrine pancreas), thyroid gland. **Secondary/associated:** liver and biliary tree, kidneys, eyes (anterior segment), inner ear (cochlea), exocrine pancreas, and rarely skull sutures, diaphragm/esophageal hiatus, heart (atrial septum), spleen, choanae.

Body systems. Endocrine (primary), hepatobiliary/digestive, renal/urinary, ophthalmic/nervous (sensory), auditory.

Tissue & cell level. - Pancreatic **beta cells** (CL:0000169) — reduced mass, impaired function/survival. - **Thyroid follicular cells** (CL:0002258) — impaired maturation. - **Renal tubular epithelium** — shortened cilia, cyst formation (CL:1000454). - **Cholangiocytes / bile-duct epithelium** — bile-duct paucity/abnormal development.

Subcellular level. **Nucleus** (GO:0005634 — GLIS3 transactivation) and the **primary cilium** (GO:0005929) are the two key compartments; GLIS3 localizes to both ([PMID: 19273592](#)). Beta-cell death proceeds through **mitochondria** (GO:0005739; intrinsic apoptosis).

"We demonstrate that Glis3 localizes to the primary cilium, suggesting that Glis3 is part of a cilium-associated signaling pathway" — [PMID: 19273592](#)

Localization / lateralization. Systemic/bilateral where paired (kidneys, eyes, cochleae). Suggested UBERON: UBERON:0000006 (islet of Langerhans), UBERON:0002046 (thyroid gland), UBERON:0002107 (liver), UBERON:0002113 (kidney), UBERON:0000970 (eye), UBERON:0001844 (cochlea).

8. Temporal Development

Onset. Congenital / neonatal. Diabetes typically presents within the first days to weeks of life (permanent neonatal diabetes, <6 months by definition), frequently with **intrauterine growth restriction**, hyperglycemia, and sometimes diabetic ketoacidosis. Congenital hypothyroidism

is present from birth (detectable on newborn screening). Onset pattern is **congenital/acute** for diabetes and **congenital/insidious** for the hepatic and renal disease, which may declare over infancy.

Progression. Diabetes is **permanent and lifelong** (not transient/self-limited). Hepatic disease can be **progressive** (hepatitis → fibrosis → cirrhosis), and renal cystic disease may progress. Overall course is chronic and, for some organs, life-limiting.

Patterns / critical periods. No spontaneous remission of the diabetes. The critical intervention windows are (i) immediate neonatal insulin initiation to control hyperglycemia and (ii) **early levothyroxine** to prevent irreversible neurodevelopmental injury from hypothyroidism. The beta-cell-development defect occurs in utero and is not reversible postnatally.

9. Inheritance and Population

Epidemiology. Neonatal diabetes mellitus overall has an incidence of **~1 in 90,000 live births** (range 1:90,000–1:160,000) [human clinical] ([PMID: 38752501](#)). GLIS3-related NDH is a **very rare recessive subset (<1% of NDM)**; on the order of ~20 families/patients are reported in the literature (Finding F013).

"These disorders are rare and the incidence is approximately 1 in 90,000 live births" —
[PMID: 38752501](#)

Inheritance. Autosomal recessive, with **high penetrance for neonatal diabetes** and **variable expressivity** for the other organ features (e.g., one compound-heterozygous patient lacked congenital hypothyroidism, [PMID: 26259131](#)). No genetic anticipation (not a repeat-expansion disorder). Germline mosaicism not specifically reported.

Consanguinity & founder effects. Recessive syndromic neonatal diabetes is strongly over-represented in **consanguineous** Middle Eastern/North African and South Asian populations, where homozygous GLIS3 large deletions and point mutations occur ([PMID: 40583116](#); [PMID: 37897565](#); [PMID: 42468610](#); [PMID: 41275391](#)).

"Monogenic diabetes is estimated to account for 1-6% of paediatric diabetes cases in primarily non-consanguineous populations, while the incidence and genetic spectrum in consanguineous regions are insufficiently defined" — [PMID: 37897565](#)

In a Saudi cohort with 81% consanguinity, autosomal-recessive syndromic and permanent neonatal diabetes predominated ([PMID: 42468610](#)); Sudanese cohorts likewise showed a predominance of syndromic recessive forms ([PMID: 41275391](#)).

Demographics. No strong sex bias (recessive). Carrier frequency is low in outbred populations, higher within consanguineous kindreds. Age distribution: affected individuals identified in the neonatal period.

10. Diagnostics

Clinical/laboratory tests. - **Glucose/insulin/C-peptide:** persistent neonatal hyperglycemia with low insulin/C-peptide (insulin-deficient), autoantibody-negative. - **Thyroid function:** low free T4 with elevated TSH (primary congenital hypothyroidism); thyroid imaging typically shows a **gland in situ** (dyshormonogenesis pattern) rather than athyreosis in most cases. - **Hepatic panel:** transaminitis/cholestasis; explant/biopsy histology shows **biliary cirrhosis / bile-duct paucity** ([PMID: 36917836](#)). - **Renal imaging (ultrasound):** cystic dysplasia / polycystic kidneys ([PMID: 26259131](#)). - **Ophthalmology:** congenital glaucoma with **high hyperopia and short axial length** ([PMID: 40583116](#)). - **Audiology:** sensorineural hearing testing. - **Pancreatic exocrine:** fecal elastase for exocrine insufficiency.

"ophthalmic assessments revealed congenital glaucoma, high hyperopia, and short axial length of the globe" — [PMID: 40583116](#)

Genetic testing (definitive). Molecular confirmation is essential and **directly influences prognosis and treatment.**

"molecular diagnosis is crucial, as it directly influences prognosis and treatment - particularly the potential responsiveness to sulfonylureas in ATP-sensitive potassium (KATP)-channel-related NDM" — [PMID: 41769619](#)

Recommended approach: - **Targeted neonatal-diabetes / monogenic-diabetes NGS gene panel** (must include GLIS3 alongside KCNJ11, ABCC8, INS, EIF2AK3, PDX1, PTF1A, GATA6, RFX6, NEUROG3, FOXP3, etc.). - **WES/WGS** for atypical/syndromic presentations and novel-variant discovery ([PMID: 38051360](#); [PMID: 42468610](#)). - **Chromosomal microarray / deletion analysis** is important because GLIS3 large multi-exon and whole-gene deletions occur ([PMID: 16715098](#); [PMID: 40583116](#)) and may be missed by SNV-only panels. - Variant interpretation per **ACMG/AMP**; functional INS-promoter luciferase assays can supply PS3-level evidence ([PMID: 38051360](#)).

Clinical criteria / differential diagnosis. Diagnosis rests on the combination of **permanent neonatal diabetes + congenital hypothyroidism** plus GLIS3 confirmation. Key differentials (Finding F014):

Gene(s)	Mechanistic class	Distinguishing feature vs GLIS3
KCNJ11 / ABCC8	KATP-channel	Sulfonylurea-responsive ; usually no CH
INS	insulin gene / ER stress	Isolated PNDM; no CH
EIF2AK3 (Wolcott-Rallison)	ER stress	Epiphyseal dysplasia, liver/renal, but not CH
GATA6 / GATA4 / PDX1	pancreatic agenesis/hypoplasia	Exocrine insufficiency + cardiac/gallbladder anomalies; labile diabetes (PMID: 41006196)
FOXP3 (IPEX)	immune dysregulation	Autoimmune enteropathy; thyroiditis (acquired, not congenital)
CISD2 (Wolfram syndrome 2)	Ca ²⁺ /ER	Later onset, optic atrophy (PMID: 40189101)
GLIS3	developmental TF	Neonatal diabetes + congenital hypothyroidism (± glaucoma, biliary cirrhosis, polycystic kidneys)

"NDM is caused by single-gene mutations that disrupt pancreatic β -cell function or development" — [PMID: 41614934](#)

Screening. Congenital hypothyroidism is captured by routine **newborn screening (TSH/T4)**; neonatal hyperglycemia prompts glucose testing. Cascade genetic testing in families follows molecular diagnosis.

11. Outcome / Prognosis

Survival/mortality. No formal survival statistics exist for this ultra-rare disorder. Prognosis is **guarded** and driven by the sum of complications: brittle insulin-dependent diabetes, and potentially **life-limiting hepatic (biliary cirrhosis) and renal disease** (PMID: 36917836). Global Glis3-knockout mice die by postnatal day ~11 from severe hyperglycemia (PMID: 31340201), underscoring the severity of complete loss; humans survive with intensive supportive care.

Morbidity/function. High: lifelong insulin dependence, neurodevelopmental risk from hypothyroidism (mitigated by early treatment), visual impairment from glaucoma, hearing loss, malnutrition from exocrine insufficiency, and organ-failure risk. QoL impact is substantial and multi-domain.

Complications & recovery. Progressive liver disease may require **liver transplantation**; renal disease may progress toward insufficiency; brittle diabetes complicates post-transplant management.

"GLIS3 mutations need to be added to the list of non-syndromic causes of bile duct paucity in the liver. Liver transplantation should be considered in patients with life-limiting complications related to liver disease" — PMID: 36917836

Prognostic factors. Severity/completeness of GLIS3 LOF, presence and progression of hepatic/renal disease, and adequacy of early metabolic and thyroid control.

12. Treatment

There is **no cure**; management is **lifelong, organ-directed supportive care** (Findings F002, F005, F010).

Pharmacotherapy. - **Insulin (exogenous)** — the cornerstone for the diabetes. GLIS3-NDH is a **beta-cell-developmental/deficiency** diabetes and is **insulin-dependent** (NCIT: C2271 Insulin). It is **NOT sulfonylurea-responsive**, in sharp contrast to KATP-channel NDM.

"patients with ABCC8 or KCNJ11 mutations treated with insulin therapy can switch to hypoglycemic sulfonylureas (SU). These drugs close the KATP channel binding the SUR1 subunit of the potassium channel and restoring insulin secretion after a meal" — [PMID: 37251668](#)

- **Levothyroxine** — lifelong thyroid-hormone replacement for congenital hypothyroidism (NCIT: C29101 Levothyroxine Sodium), started as early as possible to protect neurodevelopment.
- **Pancreatic enzyme replacement** for exocrine pancreatic insufficiency.
- **Anti-glaucoma therapy / surgery** for congenital glaucoma.

Pharmacogenomics. The single most actionable pharmacogenomic point is the **genotype-defined therapeutic split**: molecular diagnosis distinguishes sulfonylurea-responsive (KATP) from insulin-dependent (GLIS3/developmental) NDM ([PMID: 41769619](#); [PMID: 41614934](#)). Note that *monoallelic* GLIS3 T2D patients can be sulfonylurea-sensitive ([PMID: 38051360](#)) — this does **not** extend to biallelic NDH.

Surgical/interventional. Liver transplantation for life-limiting biliary cirrhosis (NCIT: C15311 Liver Transplantation); combined multi-organ transplantation (liver ± pancreas ± kidney) has been considered but performed as liver-alone in reported cases ([PMID: 36917836](#)). Glaucoma surgery as needed.

"Histology demonstrated predominantly biliary cirrhosis consistent with abnormal bile duct development" — [PMID: 36917836](#)

Advanced/experimental therapeutics. No approved gene, cell, or RNA therapy exists for GLIS3-NDH. Because the defect is developmental (failed beta-cell formation in utero), gene replacement faces the challenge that the target cells are largely absent — making **stem-cell-derived beta-cell replacement** a more plausible future avenue than in-situ gene correction. Human iPSC/hPSC CRISPR models confirm the conserved GLIS3 requirement in human pancreatic differentiation ([PMID: 27133796](#)), providing a platform for such work. No NCT-registered GLIS3-specific trials were identified.

Treatment strategy. Genotype-guided precision medicine: confirm GLIS3, commit to insulin (not a therapeutic sulfonylurea trial beyond diagnostic exclusion), replace thyroid hormone early, and institute multidisciplinary surveillance (hepatology, nephrology, ophthalmology, audiology, nutrition).

"necessitating a shift from symptomatic management to precision medicine" — [PMID: 41614934](#)

13. Prevention

- **Primary prevention:** None possible for the biallelic Mendelian disease (fully genetically determined). Prevention operates at the **reproductive/genetic-counseling** level.
- **Secondary prevention (early detection):** Newborn screening detects congenital hypothyroidism (TSH/T4); prompt recognition of neonatal hyperglycemia enables early insulin and molecular diagnosis. Early levothyroxine prevents irreversible neurodevelopmental injury.
- **Tertiary prevention:** Multidisciplinary surveillance and management of hepatic, renal, ophthalmic, and auditory complications to limit disability.
- **Genetic counseling / screening:** The key preventive strategy. Autosomal-recessive counseling for consanguineous families, **carrier/cascade testing**, and options for **prenatal diagnosis / preimplantation genetic testing** once the familial GLIS3 variant(s) are known ([PMID: 42468610](#); [PMID: 41275391](#)).
- **Immunization / public-health / environmental interventions:** Not applicable (non-infectious, non-environmental).

14. Other Species / Natural Disease

- **Taxonomy of models/orthologs:** *Mus musculus* (NCBI:txid10090), *Oryzias latipes* (medaka; NCBI:txid8090). GLIS3 is evolutionarily conserved, and its disease mechanisms are conserved across mouse, fish, and human ([PMID: 19609364](#); [PMID: 27133796](#)).
- **Orthologous genes:** mouse *Glis3*, medaka *glis3*.

- **Natural disease:** No well-documented naturally occurring companion-animal (OMIA) equivalent of GLIS3-NDH was identified; the "natural disease" model is the **medaka *pc* mutant**, which arose from a transposon insertion in *glis3* and models polycystic kidney disease ([PMID: 19609364](#)).

"the Gli-similar3 (glis3) gene was identified as the causal gene of the medaka pc mutant, a model of PKD" — [PMID: 19609364](#)

- **Comparative biology:** Glis3(zf/zf) mice recapitulate the human syndrome (diabetes + PKD), demonstrating strong cross-species conservation of both the beta-cell and ciliary-renal mechanisms.
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15. Model Organisms

Model	Type	Key phenotype / use	PMID
Global Glis3 ^{-/-} mouse	knockout, mammalian	Severe hyperglycemia; death by P~11; near-absent insulin; >90% Ngn3 reduction in embryos	31340201 , 27813676
Glis3(zf/zf) mouse	hypomorph/mutant	Diabetes + polycystic kidney disease — recapitulates human NDH	19273592
Beta-cell-specific Glis3 KO (RipCre / RosaCreERT2, TAM-inducible)	conditional	Insulin downregulation → hyperglycemia + beta-cell apoptosis; proves adult maintenance role	23197416
Glis3 ^{+/-} mouse on high-fat diet	heterozygous + environment	Diabetes from impaired beta-cell-mass expansion via Ccnd2	23197416
Thyroid-specific Glis3 KO mouse	conditional	Dysregulated thyroid gene expression; GLIS3 protein from E15.5	37461635
Medaka <i>pc</i> mutant (glis3 transposon insertion)	fish	Polycystic kidney disease from shortened renal cilia, reduced urine flow	19609364
Human iPSC/hPSC CRISPR/TALEN (GLIS3 among 8 TFs)	in vitro/cellular	Confirms conserved GLIS3 requirement in human pancreatic differentiation	27133796
INS-1E, primary rat/human islets (GLIS3 KD)	cellular	Beta-cell apoptosis via SRp55/Bim splicing	23737756

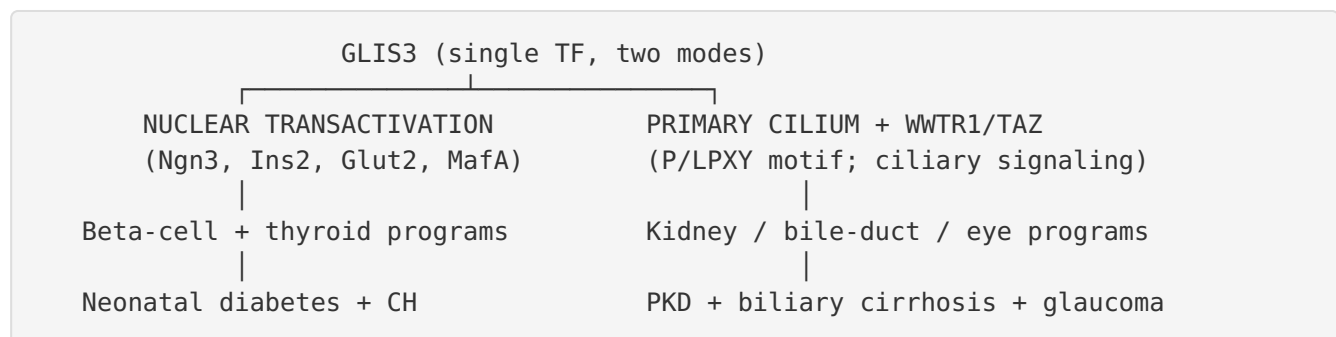
"These mice display abnormalities very similar to those of patients with neonatal diabetes and hypothyroidism syndrome, including the development of diabetes and polycystic kidney disease" — [PMID: 19273592](#)

Model characteristics / limitations. Mouse and medaka models faithfully reproduce the diabetes and renal-cystic components; conditional models successfully dissect organ-specific roles. Limitations: the ocular (high hyperopia/glaucoma) and biliary-cirrhosis phenotypes are less fully modeled, and species differences (e.g., a "potentially divergent role of NGN3 in humans and mice," [PMID: 27133796](#)) mean human iPSC systems are important complements. Resources: MGI, IMPC, ZFIN/medaka stocks; human iPSC lines from CRISPR studies.

Mechanistic Model / Interpretation

The entire NDH phenotype is best understood as the **pleiotropic failure of a single developmental transcription factor** deployed in parallel organ programs. GLIS3 is not a metabolic enzyme or an ion channel — it is an upstream *builder*. Where KATP-channel neonatal diabetes reflects a **functional** secretory defect in otherwise-formed beta cells (and is therefore drug-reversible with sulfonylureas), GLIS3-NDH reflects a **structural/developmental** deficit: the beta cells are never adequately specified (Ngn3 collapse) and the few that form are hypofunctional and apoptosis-prone (SRp55→BimS). This is the mechanistic reason the two disorders diverge so completely in treatment — a distinction that molecular diagnosis makes actionable (F005, F014).

The same logic explains the multi-organ reach. Because GLIS3 operates both as a **nuclear transactivator** and as a **cilium-associated / WWTR1-TAZ-coupled** factor, its loss simultaneously derails endocrine-pancreas development, thyroid follicular maturation, and cilium-dependent morphogenesis of kidney, biliary tree, and eye. The disorder is thus a "developmental transcription-factor syndrome" whose organ list is essentially the intersection of GLIS3's expression domains and its two molecular modes of action.



Evidence Base

PMID	Study type	Contribution
16715098	Human gene discovery	Establishes biallelic GLIS3 LOF as cause; defines core syndrome
26259131	Human case series (n=12)	Phenotypic spectrum, frequencies, variability
27813676	Mouse/gene-dosage	GLIS3→Ngn3; GWAS T1D/T2D link
31340201	Mouse/ChIP-seq	Direct binding to Ins2/Glut2/MafA; lethal global KO
23197416	Mouse conditional	Adult beta-cell maintenance; Ccnd2/diet interaction
23737756	In vitro/islets	Beta-cell apoptosis via SRp55/BimS
29246973	In vitro	SRp55 splicing network in beta cells
37461635	Mouse thyroid KO	GLIS3 in thyroid development/CH
19273592	Mouse	Cilium localization; WWTR1/TAZ; PKD
19609364	Medaka	glis3/cilia/urine-flow renal-cyst mechanism
36917836	Human explant pathology	Biliary cirrhosis/bile-duct paucity; transplant
40583116	Human siblings	Congenital glaucoma + high hyperopia; large deletion
38051360	Human resequencing + functional	Allelic dosage; ACMG functional assays; monoallelic T2D
41769619 / 37251668	Human NDM series/case	Molecular diagnosis dictates SU vs insulin
41614934	Review	NDM classification; precision medicine
38752501	Human cohort	NDM incidence ~1:90,000
37897565 / 42468610 / 41275391	Consanguineous cohorts	Recessive burden, consanguinity
27133796	hPSC editing	Conserved human GLIS3 requirement
28523428	Review	

PMID	Study type	Contribution
		GLIS3 structure/function; disease associations

Evidence source types span **human clinical** (gene discovery, case series, cohorts, explant pathology), **model organism** (mouse conditional/global KO, medaka), and **in vitro** (INS-1E, primary/human islets, iPSC), giving convergent, cross-validated support for the developmental-transcription-factor model.

Limitations and Knowledge Gaps

1. **Small n / publication bias.** Only ~20 families are reported; frequency estimates for individual organ features (glaucoma, deafness, EPI) are imprecise and likely biased toward severe, published cases.
2. **No formal natural-history or survival data.** Life expectancy, mortality rates, and validated QoL instruments have not been systematically measured for GLIS3-NDH.
3. **Genotype–phenotype correlation is incomplete.** Why some patients lack congenital hypothyroidism or have milder hepatic/renal disease (variable expressivity) is not mechanistically resolved; no proven modifier genes.
4. **Under-modeled phenotypes.** The ocular (high hyperopia) and biliary-cirrhosis components are not fully recapitulated in existing animal models.
5. **No disease-specific therapeutics.** No gene/cell/RNA therapy exists; stem-cell beta-cell replacement remains conceptual.
6. **Ontology mapping** provided here (HPO/GO/CL/UBERON/NCIT) is expert-suggested and should be curator-verified.

Proposed Follow-up Experiments / Actions

1. **Curate an international GLIS3-NDH registry** to define organ-feature frequencies, natural history, survival, and genotype–phenotype correlations with adequate power.

2. **Systematic ACMG re-classification** of all reported GLIS3 variants using the INS-promoter functional assay (PS3) framework ([PMID: 38051360](#)), including CNV/deletion detection to avoid missed large deletions.
 3. **Human iPSC-derived multi-lineage models** (beta cell, thyroid follicular, cholangiocyte, kidney organoid) from patient GLIS3 genotypes to model the ocular/biliary phenotypes currently missing in mice ([PMID: 27133796](#)).
 4. **Test whether SRp55/Bim-axis modulation** protects GLIS3-deficient human beta cells from apoptosis as a proof-of-concept therapeutic target ([PMID: 23737756](#)).
 5. **Explore stem-cell-derived beta-cell replacement** as a rational strategy given the developmental (cell-absent) nature of the defect.
 6. **Formalize a diagnostic algorithm** ensuring GLIS3 is on all neonatal-diabetes NGS panels and that neonatal diabetes + congenital hypothyroidism triggers GLIS3 deletion analysis, avoiding futile sulfonylurea trials.
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Report compiled from 14 confirmed findings and 32 reviewed publications across a 5-iteration autonomous investigation. Evidence types: human clinical, model organism, in vitro, computational.
